Kelvin Creighton

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Education

University of Alberta **2026**

Bachelor of Science in Computer Science *Edmonton, AB*

- **Focus:** Embedded Systems & Computer Architecture
- **Relevant Courses:** Computer Architecture, RISC-V Assembly, Operating Systems, Numerical Methods, Networking

Projects

Robotic Arm Simulation – CMPUT 340 | *MATLAB, Linear Algebra, Numerical Optimization* **Sep 2024 – Dec 2024**

- Developed an interactive 2D robotic arm simulation in MATLAB to calculate the joint angles needed to move the arm to a target position.
- Derived and implemented an analytical Jacobian, along with a finite-difference approximation to evaluate how changes in joint angles affect the arm's position.
- Implemented Newton's method and Broyden's method to iteratively solve for joint angles, comparing an exact Jacobian approach with a lower-computation quasi-Newton approach.

Dremel 3D45 Control System Replacement | *Arduino Mega 2560, RAMPS 1.4, Marlin, C++* **May 2026 – Present**

- Replaced the failed proprietary controller of a Dremel 3D45 3D printer with an Arduino Mega 2560 and RAMPS 1.4 control system.
- Configured, compiled, and flashed Marlin firmware for the replacement controller.
- Rewired stepper motors, endstops, heater, and temperature-sensing circuitry to interface with the new control system.
- Diagnosed hardware and wiring issues using a multimeter, including continuity, resistance, and short-circuit checks during system bring-up.

Linux AI Homelab Cluster | *Debian, Linux, Networking, Tailscale* **May 2026 – Present**

- Built and maintained a multi-node Debian Linux cluster using repurposed hardware for development and self-hosted workloads.
- Configured SSH administration, firewall policies, Tailscale networking, and automated filesystem mounting across nodes.

Experience

Technology Specialist Co-op **May 2024 – Sep 2024**

DiscoverE — University of Alberta *Edmonton, AB*

- Provided technical support across 20+ classrooms and office environments, troubleshooting hardware, software, and A/V issues across 20+ devices.
- Simplified an employee sign-in/out system used by 20+ staff, removing 80% of redundant functions and scripts and reducing sign-in errors.
- Developed an internal inventory management system using Google Sheets and Apps Script, implementing role-based access controls and data validation.
- Identified a gap in organizational data resilience and implemented secure on-premises backups to improve disaster recovery.

Leadership

Chapter Executive – OWASP **Jan 2026 – Present**

Open Worldwide Application Security Project (OWASP) — Edmonton Chapter *Edmonton, AB*

- Supported technical event infrastructure and resolved HDMI compatibility and audio issues during presentations.
- Organized a Secure Coding Tournament for 50+ participants involving vulnerability identification and remediation.

VP Events – C.H.A.D.S. **Sep 2024 – Sep 2026**

Cybersecurity, Hacking, And Digital Security club (CHADS) — University of Alberta *Edmonton, AB*

- Configured Linux-based CTF infrastructure on Linode and containerized 25+ challenges using Docker and Python.
- Organized technical events and competitions serving 200+ participants.

Skills

Technical: C, C++, Python, Bash, JavaScript, SQL, Marlin, Hardware Troubleshooting, Electronics Debugging, Linux, Git, Docker, Networking, CI/CD, MATLAB, Arduino IDE, RTOS, I2C, STM32

Professional: Leadership, Communication, Event Organization, Technical Documentation & Reporting, Troubleshooting, Problem Solving, Cross-functional Collaboration